Andrew Causer

Senior Bioinformatician

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Brisbane, Australia

PROFILE

Passionate Data Analyst and Bioinformatician with expertise in spatial multi-omics, big data analytics, machine learning and clinical biomedical research. Acquired from my education and previous roles, I bring a diverse skill-set. This includes a strong proficiency in various coding languages, team leadership and mentoring skills, and expertise in processing and analysing clinical oncology samples. I am adept at managing projects of high-importance independently and effectively, as well as collaborating as part of a productive team. My dedication to continuous learning and growth fuels my professional development.

EXPERIENCE

Senior Bioinformatician

Institute of Molecular Biology (UQ)

08/2023 - Present

- Leveraging spatial-omic technologies and deep learning methodologies to analyse various complex diseases, including unravelling intricate biological mechanisms associated with diverse tumour micro-environments
- Strong skills in manipulating Seurat/scanpy data objects along with H&E image processing for applying various deep learning algorithms
- Analysing datasets including spatial proteomics (Codex), spatial transcriptomics (Visium/Xenium/CosMx/STOmics), spatial metabolomics (MALDI-TOF) and various bulk data
- Organising group meetings and events, assisting with staff onboarding, learning and development, running teaching workshops at QIMR and UQ
- Presented at various conferences (10X Genomics Brisbane) and institute meetings

Machine Learning Researcher

Max Kelsen

03/2023 - 04/2023

- Further applied and developed skills in big dataset manipulation and analysis
- Used R and Python to computationally analyze over 15,000 tumour cases spread across over 10 different datasets
- Strong understanding of handling and manipulating Genomic Data Commons data
- Acquired skills in Google Cloud computing and GitHub version control
- Effectively worked to deliver tasks and results by strict deadlines

Clinical Laboratory Scientist

Sullivan Nicolaides Pathology

01/2021 - 03/2023

- Developed strong histology laboratory skills such as embedding, microtomy and H&E staining
- Handle a diverse range of complex human specimen
- Work efficiently and produce high quality samples in high pace work environment
- Skills and Knowledge surrounding the use and maintenance of histology tissue processing machines
- Presented for the laboratory staff education program

Clinical Laboratory Assistant

Sullivan Nicolaides Pathology

03/2020 - 01/2021

Retail Worker

Woolworths

02/2017 - Present

- Clear communication with customers to meet consumer satisfaction, achieving sales goals and problem solving in a fast-paced environment
- Communicating with team members, taking responsibilities for tasks and achieving them in an efficient and effective manner
- Experience in food safety, requiring a high level of commitment and care when completing tasks in the bakery and deli departments

FIND ME ONLINE

GitHub
github.com/agc888

Linkedin
linkedin.com/in/andrewgcauser

INDUSTRY EXPERTISE

- **Bioinformatics**
- **R Programming**
- **Python Programing**
- **Advanced Statistical Analysis**
- **Spatial Multi-Omic Analysis**
- **Clinical Research**
- **Data Analysis/Problem Solving**
- **Machine Learning**
- **R Package Design**
- **Java Script**
- **Google Cloud/AWS**
- **Adobe Illustrator/Photoshop**
- **H&E Image Manipulation**
- **Data Presentation**
- **Version Control**
- **Customer Service**
- **Biomedical Science**
- **Oncology**
- **Genomics**

EDUCATION

Master of Bioinformatics, Research Extensive

University of Queensland

06/2021 - 11/2022

GPA **6.94** / 7.0

- **Publication in Nature Precision Oncology IF: 7.9** (2022)
- Advanced genomic data analysis through computational problem solving (Python and R)
- Thesis project involves the integration of multiomics datasets from human head and neck cancer patients in-order to distinguish heterogeneity between patients, assess statistical performance of multiomics integration, and identify drug targets for immunotherapies
- Presented at the CellOmics Innovation Forum and the MultiOmics 2022 Conference (Poster Presentation)
- Proficient in clearly and concisely presenting data and results for publication, presentations and general meetings

Graduate Certificate, Bioinformatics

University of Queensland

01/2020 - 06/2021

GPA **7.0** / 7.0

- Analysed biological/genetic data through computational problem solving
- Developed various programs with strong programming skills in Python and R
- Basic understanding of database management systems (SQL)
- Proficient in Microsoft Office programs (Excel, Word, PowerPoint)

Bachelor of Science, Extended Major in Biomedical Science

University of Queensland

02/2016 - 11/2019

GPA **6.4** / 7.0

- **Honours (First Class)**
- Honours Project: Developing a clinically relevant mouse model of high-level spinal cord injury to study circadian metabolism and immune dysfunction
- Specialised in pathophysiology, developmental biology and neuroscience
- Received multiple Dean's Commendation for Academic Excellence (3)
- Third year developmental biology prize (2019)

Deep Learning Specialisation

Coursera: DeepLearning.AI

07/2023 - 08/2023

- **Offical Certification**
- Understand the statistical and mathematical nuances of neural networks and deep learning
- Practical application of generating, testing and fine-tuning various types of neural networks and parameters
- Coding skills in Python and Tenserflow

REFERENCES

Dr Quan Nguyen

Institute of Molecular Bioscience, UQ
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Dr Scott Allen

Max Kelsen, Senior Machine Learning Researcher Email: scottallen@outlook.com
Mobile: +61 492 883 786

Katie Ziegenfusz

Sullivan Nicolaides Pathology, Histology
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Phone: 07 3878 0944

ADDITIONAL SKILLS

Figure Design

Generated Figures for Nature Papers

Logo Design

Helped design and generate the logo for GenomicsMachineLearning Research Group